Influence of Oxygen on the Optical and Electrical Properties of Magnetron-Sputtered Indium Tin Oxide Thin Films at Ambient Temperature

Alex J. Robb,[1] Zachary A. Duca,[1] Nasiba White[1], Patrick Woodell[2], and Patrick A. Ward[1,*]

[1]Materials Technology and Energy Directorate, Savannah River National Laboratory, 301 Gateway Drive, Aiken, SC, 29803, USA

[2]Applied Research Center, 301 Gateway Drive, Aiken, SC, 29803, USA

*Corresponding Author: patrick.ward@srnl.doe.gov

Abstract

Indium Tin Oxide (ITO) thin films are commonly used as transparent conductive layers in the production of solar cells. ITO films are typically heat-treated at temperatures exceeding 200 °C after magnetron sputtering to optimize the physical, optical, and electrical characteristics for photovoltaic devices. This high temperature heating procedure greatly limits the use of potential substrates, particularly those comprised of low-temperature-rated polymers. Herein, we examine the physical, optical, and electrical transport effects of oxygen:argon plasmas during radio frequency magnetron sputtering of ITO on glass and flexible polymer surfaces without subsequent heat treatment. ITO thin films were characterized via sheet resistance testing, UV-Vis spectroscopy, X-ray diffraction, and scanning electron microscopy imaging. Sheet resistance measurements revealed an increase in surface resistance with increasing oxygen concentration during sputtering. UV-Vis optical transmittance measurements demonstrated an improvement in optical transmittance with an increase in oxygen concentration reaching a maximum transmittance at an oxygen concentration of 1-2 %. Most importantly, it was discovered that producing bilayers of ITO with pure argon sputtered plasmas followed by oxygen:argon plasmas

produced ITO thin films with suitable resistance and transmittance without the need for high temperature post annealing. These results highlight a methodology for the deposition of ITO thin films on temperature sensitive substrates for flexible photovoltaics and low-cost module production.

Key Words: *Indium Tin Oxide, Solar Cells, Radio-Frequency Magnetron Sputtering, photovoltaics, flexible substrates*

1. Introduction

Transparent conductive oxides (TCO) have been extensively studied for a wide variety of applications including electrical contacts [1, 2], photochromic displays [3], and various optoelectronic and photovoltaic devices [4]. TCOs are a critical component of photovoltaics due to their high electrical conductivity and optical transmittance properties. While a number of TCOs are known, such as aluminum doped zinc oxide, fluorine doped tin oxide, and magnesium doped zinc oxide, indium doped tin oxide (ITO) is often used due to excellent electrical and optical properties as well as its ease and flexibility of production [5, 6]. ITO films can be created through a variety of means such as thermal evaporation [7], chemical vapor deposition [8], sol–gel methods [9], spray pyrolysis [10], atomic layer deposition [11], and most notably in industry through radio frequency (RF) or direct current magnetron sputtering [12, 13].

Magnetron sputtering of ITO has favorable outcomes, such as high reproducibility, fast deposition times, large surface areas for coating, and the ability to control the parameters of deposition, such as power, gas flow and composition, chamber pressure, and substrate temperature. However, common sputtering temperatures during deposition are >200 °C and regardless of sputtering, ITO films are typically annealed post-sputtering at >400 °C to produce films with enhanced transparency and optimal electrical resistance. This method is not suitable for photovoltaic applications in which organic materials or flexible polymeric substrates such as polyethylene terephthalate (PET) are used due to temperature stability of the substrate. This being said, several instances of magnetron sputtering on PET without heating have been reported for applications in flexible electronics [14-16].

It is well-known that oxygen defects in ITO films influence the thin film transmittance and resistivity [17]. Herein, we present a pathway for enhancing the electronic and optical

properties of room temperature, RF magnetron sputtered ITO by strategically controlling the oxygen concentration of the argon plasma during deposition. This was done through creating bilayer films, in which an initial layer, without oxygen present during the sputtering process, was deposited to create a pathway for electrons to travel through the film, while an oxygen containing layer allowed for increased visible light transmittance through the film.

2. Experimental Details

2.1 Reagents

All reagents were used as purchased except where noted. Oxygen (99.9995 %) and argon (99.995 %) were purchased from NexAir (Augusta, GA, USA). Quartz slides (1 in. x 1 in. x 0.062 in.) were purchased from Technical Glass Products Inc. (Painesville, OH, USA). Polyethylene terephthalate glycol (PETG) sheets were purchased from McMaster-Carr (Cleveland, OH, USA). Indium Tin Oxide magnetron sputtering targets were purchased from Kurt J. Lesker (Jefferson Hills, PA, USA). Hydrochloric Acid (HCl) and acetone was used as purchased (Sigma). Deionized (DI) water was generated from a MilliporeSigma MilliQ water purification system with a measured resistance greater than 18 MΩ•cm.

2.2 Cleaning Procedures for Quartz Slides and PETG Sheets

Quartz slides (2.5 × 2.5 cm) were sonicated for 30 minutes in 2 M HCl, rinsed with DI water, then sonicated for 10 minutes in DI water and rinsed in acetone. The slides were dried for 30 minutes in an oven at 100 °C and UV/ozone cleaned for 10 minutes immediately before thin film deposition. PETG sheets were cut into 2.5 × 2.5 cm slides and the slides were sonicated for 30 minutes in DI water and then dried for 30 minutes in an oven at 100 °C. Sheets were UV/ozone cleaned for 10 minutes immediately before thin film deposition.

2.3 RF Magnetron Sputtering

ITO film deposition was performed via magnetron sputtering in a home-built Kurt J. Lesker sputtering apparatus utilizing a RF power supply, a Taurus gun, and an ITO target (Kurt J. Lesker, 2.00 inch diameter, 99.99 % purity). The gun was held at a distance of 10 cm directly above rotational stage, which was spun at a rate of 13 rotations per minute. Previous studies have shown that sputtering in an oxygen containing environment can affect the deposition rate of ITO [18-20], which is due to oxidation of the sputtering target during deposition. Therefore, to avoid oxidation of the target and a change in deposition rate, the target was ramped up to power (60 W) at a rate of 4 W/minute and was then maintained at that power for 15 minutes in an oxygen free environment to activate the target. After activation of the target surface, oxygen exposure was limited to scenarios where a specific oxygen concentration was being evaluated for sputtering. The composition of gas flowing into the chamber was controlled via two flow controllers, one controlling the amount of pure argon and the other controlling a 9:1 oxygen:argon mixture. To achieve the intended percent of oxygen (0-8 %) in the inlet flow to the sputtering chamber, the operational pressure of the chamber was maintained at 0.47 Pa and the total flow rate of the two gases was maintained at 170 standard cubic centimeters per minute (sccm), but the amounts of either gas (argon or argon:oxygen) were modified and the conditions are shown in **Table 1**. For example, to create a film in which oxygen made up 8 % of the total inlet flow into the sputtering chamber while sputtering, then the flow of argon would be set to 34 sccm and the flow of the 9:1 argon oxygen mix would be set to 136 sccm. Since the total flow into the chamber was maintained at 170 sccm, the argon:oxygen mixture comprised 80 % of the total gas flow. Given a gas mixture ratio of 9:1, this provides an 8 % oxygen composition in argon.

Table 1. Sputtering conditions for all mono- and bilayer films produced in this work.

Sample Name	**Step 1:** ITO deposition time with pure Argon (min)	**Step 2:** ITO deposition time with the Ar:O_2 mix (min)	**Ar:O_2 Mix**				
			Pure Argon Flow (sccm)	(9:1) Ar:O_2 Mix Flow		Total Gas Flow Ar + Ar:O_2 (sccm)	O_2 Flow (%)
				Total (sccm)	O_2 Composition (sccm)		
Ar:O_2 8%	0	30	34	136	13.6	170	8
Ar:O_2 4%	0	30	102	68	6.8	170	4
Ar:O_2 2%	0	30	136	34	3.4	170	2
Ar:O_2 1%	0	30	153	17	1.7	170	1
Ar:O_2 0.5%	0	30	161.5	8.5	0.85	170	0.5
Ar:O_2 0.25%	0	30	165.75	4.25	0.425	170	0.25
Ar 0%	30	0	170	0	0	170	0
Ar + Ar:O_2 0.25%	5	25	165.75	4.25	0.425	170	0.25
Ar + Ar:O_2 0.5%	5	25	161.5	8.5	0.85	170	0.5
Ar + Ar:O_2 1%	5	25	153	17	1.7	170	1
Ar + Ar:O_2 2%	5	25	136	34	3.4	170	2
Ar + Ar:O_2 4%	5	25	102	68	6.8	170	4
Ar + Ar:O_2 8%	5	25	34	136	13.6	170	8

2.4 Magnetron Sputtering of ITO Films onto Substrates

Cleaned substrates were placed onto a rotational stage in a vacuum chamber and evacuated to low pressures (1.3×10^{-5} Pa) to remove ambient oxygen and contaminants from the chamber. Pure argon was then flowed into the chamber until the system reached the operating pressure of 0.47 Pa and the chamber was allowed to purge at this pressure for 30 minutes prior to sputtering. Sputtering of ITO was performed at a power of 60 watts RF for a total deposition time of 30 minutes for all samples. The percentage of oxygen present in the total argon:oxygen gas mixture that was flowed into the chamber during this 30-minute deposition period was varied to create either monolayer or bilayer films with different oxygen compositions. Monolayer films (**Ar:O_2**) were produced by allowing a fixed percentage of oxygen (0-8 %) to be present in the total gas flow into the chamber throughout the entire 30-minute sputter time. Bilayer films (**Ar + Ar:O_2**) were produced by sputtering in the absence of oxygen (pure argon) for the first 5-minutes followed by 25 minutes of deposition with a fixed percentage (0-8 %) of oxygen in the total gas flow into the chamber (**Scheme 1**).

2.5 Sheet Resistance Measurements of ITO Thin Films

Sheet resistance of the sputtered substrates was determined using a Jandel RM3-AR four-point probe test unit utilizing a 1 μA compliance current, sampling the center of the ITO films. Measurements were performed immediately after sputtering to minimize any potential influence of long-term ambient oxygen exposure on the sample film properties.

2.6 UV-Vis Measurements of ITO Thin Films

UV-Vis transmittance measurements were performed using a PerkinElmer Lambda 45 dual beam UV-Vis spectrometer. The instrument was blanked to air and a cleaned, non-coated quartz or PETG substrate was used as the reference. Scans were performed from 300-900 nm at a rate of 8 nm/second immediately after sputtering to minimize ambient oxygen interactions with the sample films.

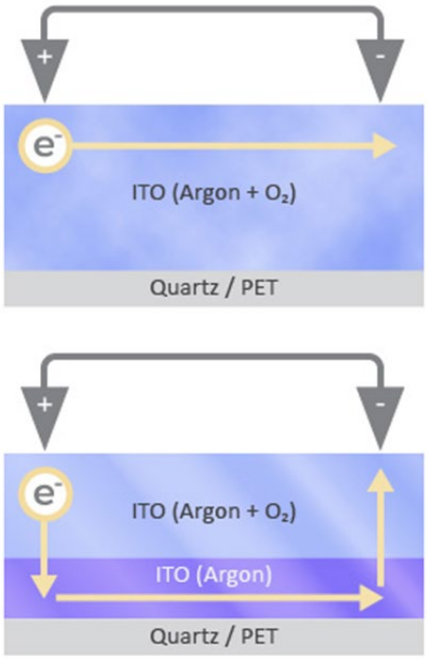
2.7 Microscopy

Scanning Electron Microscopy (SEM) images were taken on a Hitachi Su8230 Field Emission Scanning Electron Microscope. All SEM images were obtained at accelerating voltages of 3 keV and working Distance of 4-5 mm were used.

2.8 X-ray Diffraction

Powder X-ray diffraction (XRD) patterns were measured on a PANalytical X'pert Pro with Cu Kα radiation in a Bragg-Brentano parafocusing geometry.

3. Results and Discussion



Scheme 1. Cartoon representation of the proposed mono- (top) and bilayer (bottom) films produced in this study

3.1 Sheet Resistance

Evaluation of the sheet resistance of films sputtered under various concentrations of oxygen demonstrate a clear increase in the resistance of the thin film with increasing oxygen concentration as shown in **Figure 1**. While the ITO thin films produced with a constant ratio of $Ar:O_2$ during sputtering demonstrated consistent reduction in electrical conductivity with increasing oxygen concentration, films which were produced with a pure argon atmosphere followed by an $Ar:O_2$ atmosphere demonstrate a minimal increase in measured sheet resistance as illustrated in **Figure 1**.

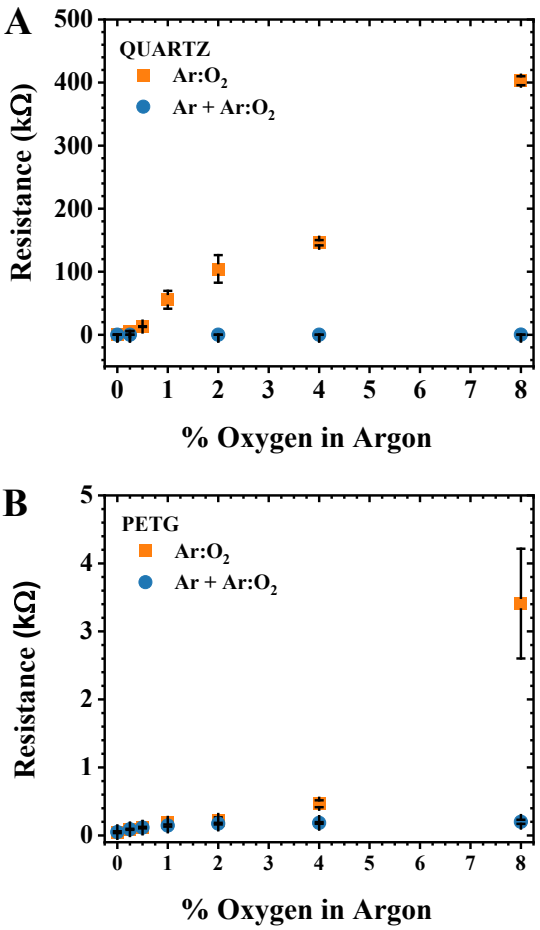


Figure 1. Average sheet resistance of Quartz (A) and PETG (B) slides containing ITO deposited under varying oxygen flow compositions without ($Ar:O_2$) or with ($Ar + Ar:O_2$) an initial ITO layer deposited under pure argon. (Error values were determined from the standard error from a set of 4 measurements.)

On quartz, the addition of an ITO layer sputtered under pure argon followed by $Ar:O_2$ resulted in a decrease in the sheet resistance. For the 0.25 % $Ar + Ar:O_2$ sample, there was a decrease in resistance from 5.93 kΩ to 191 Ω, a decrease of ~ 5.74 kΩ (96 %). For the 4 % $Ar + Ar:O_2$ sample, there was a decrease in resistance from 0.146 MΩ to 194 Ω, a ~ 145 kΩ (99 %) reduction. To demonstrate the feasibility of this methodology on a flexible substrate, PETG sheets were sputtered under the same conditions. The results illustrate a similar trend, but less

drastic differences in resistance are observed. A decrease of ~ 4 Ω (4 %) for the 0.25 % Ar + Ar:O_2 sample and ~ 282 Ω (60 %) for the 4 % Ar + Ar:O_2 sample were observed when compared to the bilayer containing film. The dramatic difference in resistance between the quartz and PETG substrate is likely due to the packing and morphology differences of the film growth on the different substrates. ITO films grown on PETG were found to have a more compact film than those sputtered on quartz.

3.2 UV-Vis

UV-Vis measurements were performed to determine the percent transmittance of the films at different wavelengths and compare their feasibility as transparent conducting layers in photovoltaic applications. ITO typically absorbs from 200-400 nm in the visible and has broader absorbance in wavelengths greater than 1000 nm leaving a transmission window from ~400-900 nm. For all samples, there was an observable hypochromic shift in the absorbance of the films with the increase of O_2 concentration during the sputtering process (**Figure 2**). An increase in the hypochromic shift was observed for all samples with increased oxygen concentration in the sputtering plasma. Also, an initial increase in the film's transmittance in the 350-500 nm range was observed until ~1-2 % O_2 composition was reached.

When comparing the samples without and with a 5-minute pure argon deposition period on quartz, a decrease in the transmittance of the films upon the addition of that base argon layer was observed (**Figure 2A, C**). However, on PETG the transmittance of the films with and without the base layer remained relatively similar (**Figure 2B, D**).

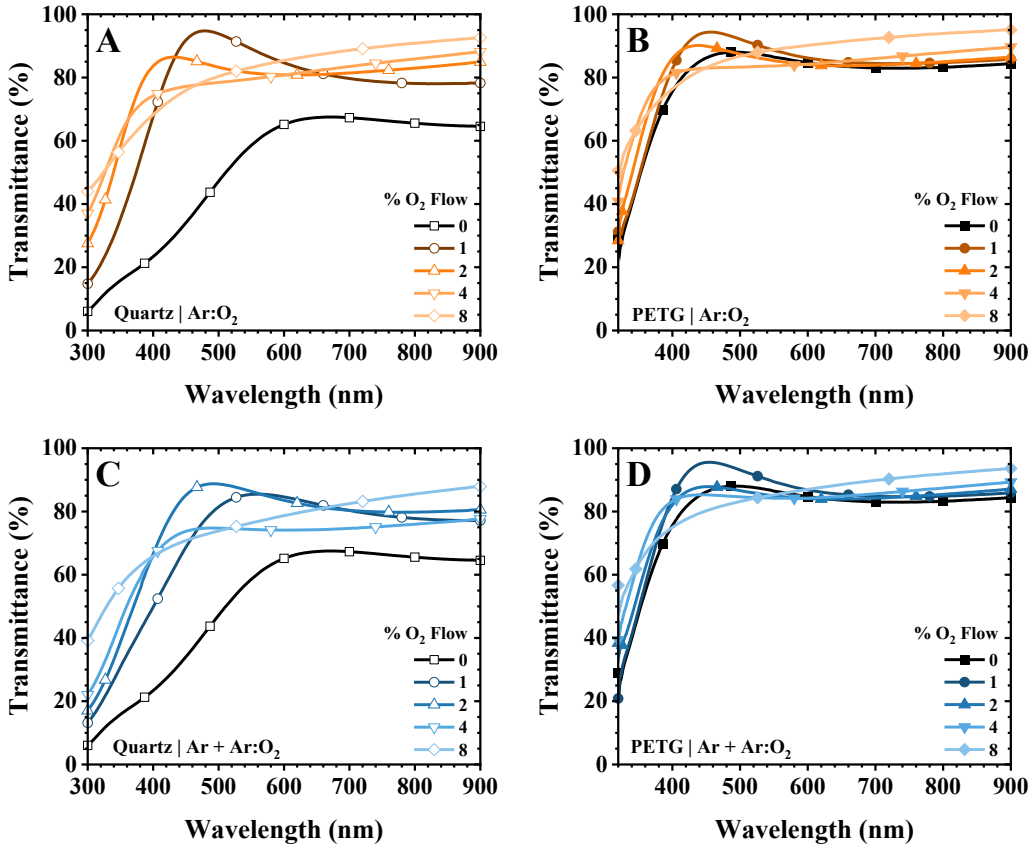


Figure 2. UV-Vis transmission measurements of Quartz and PETG slides containing ITO deposited under varying oxygen flow compositions without (A)(B) or with (C)(D) an initial ITO layer deposited under pure argon.

Table 2. Average percent transmission in the visible wavelength range (350–750 nm) for each sample.

	$Ar:O_2$					$Ar + Ar:O_2$				
% Oxygen	0%	1%	2%	4%	8%	0%	1%	2%	4%	8%
Average %T from 350-750 nm on Quartz	50.3	80.6	81.3	78.8	80.3	50.3	73.2	78.9	71.6	74.9
Average %T from 350-750 nm on PETG	81.8	85.2	84.3	83.1	86.0	81.8	86.1	83.2	83.7	83.4

3.3 Film Morphology

SEM imaging of the ITO thin films highlighted a consistent morphology across all surfaces (**Figure 3**), suggesting that resistance and transmittance of samples was not influenced by surface defects and significant changes in morphology. RF sputtered ITO films with and

without oxygen consist of amorphous nanoscale grains, with a minor increase in grain size in the presence of oxygen during sputtering as observed in previous work [21]. Cross-sectional SEM images of the ITO **(Figure 3D**) films reveal a tightly packed structure with the thickness of the deposited ITO layers ca. 150-180 nm. The ITO bilayer thin film with 2% oxygen (**Figure 3B**) appears to be more densely packed and interestingly has more crystalline content than the amorphous ITO thin films produced under steady state argon:oxygen concentrations (**Figure 4A**). High temperature annealing of these films resulted in an increase in the overall crystallinity (**Figure 4B**) as anticipated. Whereas, crystallinity was only observed on low oxygen concentration bilayer films when room temperature sputtered ITO thin films were prepared without subsequent annealing.

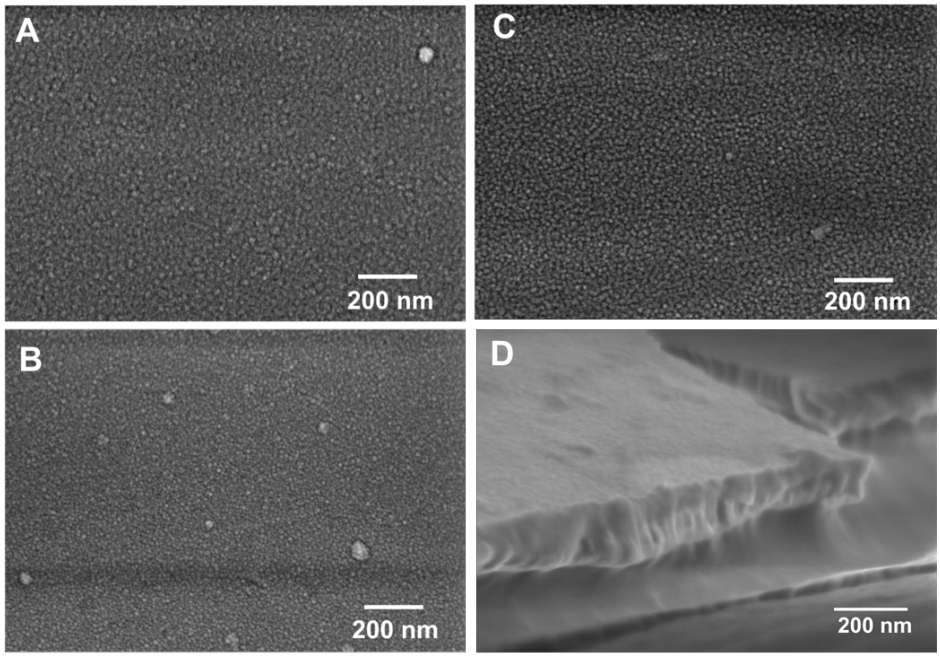


Figure 3. SEM images of ITO samples fabricated using RF-magnetron sputtering on Quartz at various O_2 concentrations: A) Ar (0 % O_2), B) Ar + Ar:O_2 (2% O_2), C) Ar + Ar:O_2 (8 % O_2) and D) cross sectional SEM of the Ar + Ar:O_2 (8 % O_2) film on PETG.

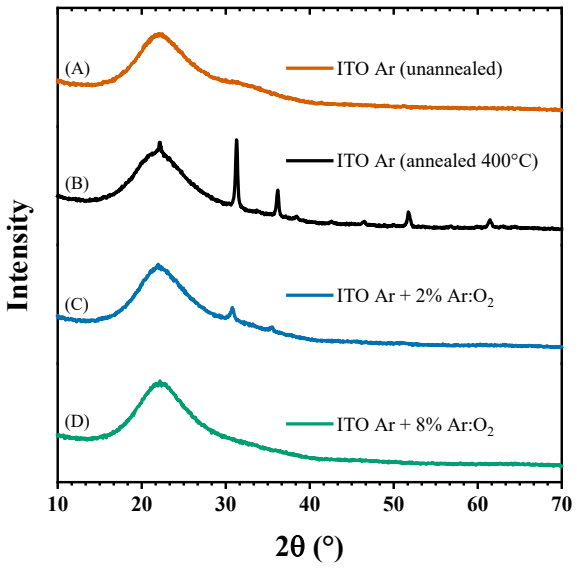


Figure 4: XRD patterns of ITO thin films on quartz substrates sputtered under A) pure argon (Ar, 0 % O_2) B) pure argon (Ar, 0 % O_2) followed by annealing in air at 400 °C for 4 hours C) pure argon for 5 minutes then argon:oxygen with 2 % oxygen for 25 minutes (Ar + Ar:O_2 2 %) and D) pure argon followed by argon:oxygen with 8 % oxygen minutes (Ar + Ar:O_2 8 %) for 25 minutes

4. Conclusions

In this work, we sought to maximize the electrical conductivity and optical transmittance of room temperature, RF magnetron sputtered ITO by strategically controlling the oxygen composition of the gas flow during deposition. It was found that the formation of a bilayer of ITO sputtered under pure argon and ITO sputtered in an argon: oxygen atmosphere resulted in enhanced electrical conductivity with a minimal impact on optical transmittance. This result is believed to be due to electrons taking the path of least resistance through the bottom, more conducting, pure argon sputtered layer while the top oxygen sputtered layer provides higher optical transparency. We found significant decreases in the sheet resistance of the films upon the incorporation on an initial oxygen-free ITO layer on both substrates (quartz and PETG). We also found that increasing the oxygen concentration during magnetron sputtering leads to an increase

in the transparency of the films, with a maximum transparency achieved at 2 % oxygen and then decreasing thereafter. The addition of an initial ITO layer sputtered under Ar did not have a significant decrease in the observed transmittance of the films when compared to a film without the initial Ar ITO layer. These results demonstrate the potential for the deposition of ITO thin films with low resistivities on flexible and low-cost substrates without the need for post annealing at high temperatures.

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References

(1) Leterrier, Y.; Médico, L.; Demarco, F.; Månson, J. A. E.; Betz, U.; Escolà, M. F.; Kharrazi Olsson, M.; Atamny, F. Mechanical integrity of transparent conductive oxide films for flexible polymer-based displays. *Thin Solid Films* **2004**, *460* (1), 156-166.
(2) Kikuchi, N.; Kusano, E.; Kishio, E.; Kinbara, A. Electrical and mechanical properties of SnO2:Nb films for touch screens. *Vacuum* **2002**, *66* (3), 365-371.
(3) Wang, Q.; Diez-Cabanes, V.; Dell'Elce, S.; Liscio, A.; Kobin, B.; Li, H.; Brédas, J.-L.; Hecht, S.; Palermo, V.; List-Kratochvil, E. J. W.; et al. Dynamically Switching the Electronic and Electrostatic Properties of Indium–Tin Oxide Electrodes with Photochromic Monolayers: Toward Photoswitchable Optoelectronic Devices. *ACS Applied Nano Materials* **2019**, *2* (2), 1102-1110.
(4) Liu, H.; Avrutin, V.; Izyumskaya, N.; Özgür, Ü.; Morkoç, H. Transparent conducting oxides for electrode applications in light emitting and absorbing devices. *Superlattices and Microstructures* **2010**, *48* (5), 458-484.
(5) Calnan, S.; Tiwari, A. N. High mobility transparent conducting oxides for thin film solar cells. *Thin Solid Films* **2010**, *518* (7), 1839-1849.

(6) Hotovy, I.; Pezoldt, J.; Kadlecikova, M.; Kups, T.; Spiess, L.; Breza, J.; Sakalauskas, E.; Goldhahn, R.; Rehacek, V. Structural characterization of sputtered indium oxide films deposited at room temperature. *Thin Solid Films* **2010**, *518* (16), 4508-4511.
(7) Du, J.; Chen, X.-l.; Liu, C.-c.; Ni, J.; Hou, G.-f.; Zhao, Y.; Zhang, X.-d. Highly transparent and conductive indium tin oxide thin films for solar cells grown by reactive thermal evaporation at low temperature. *Applied Physics A* **2014**, *117* (2), 815-822.
(8) Maruyama, T.; Fukui, K. Indium-tin oxide thin films prepared by chemical vapor deposition. *Journal of Applied Physics* **1991**, *70* (7), 3848-3851.
(9) Cho, H.; Yun, Y.-H. Characterization of indium tin oxide (ITO) thin films prepared by a sol–gel spin coating process. *Ceramics International* **2011**, *37* (2), 615-619.
(10) Vasu, V.; Subrahmanyam, A. Photovoltaic properties of indium tin oxide (ITO)/silicon junctions prepared by spray pyrolysis-dependence on oxidation time. *Semiconductor science and technology* **1992**, *7* (3), 320.
(11) Mane, A. U.; Allen, A. J.; Kanjolia, R. K.; Elam, J. W. Indium Oxide Thin Films by Atomic Layer Deposition Using Trimethylindium and Ozone. *The Journal of Physical Chemistry C* **2016**, *120* (18), 9874-9883.
(12) Lee, H.-C.; Seo, J.-Y.; Choi, Y.-W.; Lee, D.-W. The growth of indium-tin-oxide thin films on glass substrates using DC reactive magnetron sputtering. *Vacuum* **2003**, *72* (3), 269-276.
(13) Park, J.-O.; Lee, J.-H.; Kim, J.-J.; Cho, S.-H.; Cho, Y. K. Crystallization of indium tin oxide thin films prepared by RF-magnetron sputtering without external heating. *Thin Solid Films* **2005**, *474* (1), 127-132.
(14) Ahn, M. H.; Cho, E. S.; Kwon, S. J. Characteristics of ITO-resistive touch film deposited on a PET substrate by in-line DC magnetron sputtering. *Vacuum* **2014**, *101*, 221-227.
(15) Tseng, K.-S.; Lo, Y.-L. Effects of cumulative ion bombardment on ITO films deposited on PET and Si substrates by DC magnetron sputtering. *Opt. Mater. Express* **2014**, *4* (4), 764-775.
(16) Lee, J. R.; Lee, D. Y.; Kim, D. G.; Lee, G. H.; Kim, Y. D.; Song, P. K. Characteristics of ITO films deposited on a PET substrate under various deposition conditions. *Metals and Materials International* **2008**, *14* (6), 745-751.
(17) Chen, A.; Zhu, K.; Zhong, H.; Shao, Q.; Ge, G. A new investigation of oxygen flow influence on ITO thin films by magnetron sputtering. *Solar Energy Materials and Solar Cells* **2014**, *120*, 157-162.
(18) Choi, C. G.; No, K.; Lee, W.-J.; Kim, H.-G.; Jung, S. O.; Lee, W. J.; Kim, W. S.; Kim, S. J.; Yoon, C. Effects of oxygen partial pressure on the microstructure and electrical properties of indium tin oxide film prepared by d.c. magnetron sputtering. *Thin Solid Films* **1995**, *258* (1), 274-278.
(19) Das, R.; Adhikary, K.; Ray, S. The role of oxygen and hydrogen partial pressures on structural and optical properties of ITO films deposited by reactive rf-magnetron sputtering. *Applied Surface Science* **2007**, *253* (14), 6068-6073.
(20) Kerkache, L.; Layadi, A.; Mosser, A. Effect of oxygen partial pressure on the structural and optical properties of dc sputtered ITO thin films. *Journal of Alloys and Compounds* **2009**, *485* (1), 46-50.
(21) Krasilnikova Sytchkova, A.; Grilli, M. L.; Boycheva, S.; Piegari, A. Optical, electrical, structural and microstructural characteristics of rf sputtered ITO films developed for art protection coatings. *Applied Physics A* **2007**, *89* (1), 63-72.